

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

PARSE BIOSCIENCES, INC.,
Petitioner,

v.

10X GENOMICS, INC.,
Patent Owner.

IPR2023-00876
Patent 10,155,981 B2

Before MICHAEL J. FITZPATRICK, SHERIDAN K. SNEDDEN, and
DEVON ZASTROW NEWMAN, *Administrative Patent Judges*.

FITZPATRICK, *Administrative Patent Judge*.

JUDGMENT
Determining All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

A. *Background and Summary*

Petitioner, Parse Biosciences, Inc., filed a Petition to institute an *inter partes* review of all claims, namely claims 1–6, of U.S. Patent No. 10,155,981 B2 (“the ’981 patent”) pursuant to 35 U.S.C. § 311(a). Paper 1 (“Pet.”). Patent Owner, 10X Genomics, Inc., filed a Preliminary Response pursuant to 37 C.F.R. § 42.107(b). Paper 6 (“Prelim. Resp.”). We granted the Petition and instituted an *inter partes* review. Paper 7 (“Inst. Dec.”).

During the review, Patent Owner filed a Patent Owner Response to the Petition (Paper 13 (“PO Resp.”)), Petitioner filed a Reply (Paper 22 (“Pet. Reply”)), and Patent Owner filed a Sur-Reply (Paper 26 (“PO Sur-Reply”). Additionally, each party filed a motion to exclude. Paper 34 (“Pet. Mot. To Exclude”); Paper 35 (“PO Mot. To Exclude”).¹

An oral hearing was held July 24, 2024. A transcript of the hearing is of record in this case. Paper 46 (“Tr.”).

We have jurisdiction under 35 U.S.C. § 6, and this Final Written Decision, issued pursuant to 35 U.S.C. § 318(a), addresses issues and arguments raised during the trial. For the reasons discussed below, we determine that Petitioner has proven, by a preponderance of the evidence, that claims 1–6 of the ’981 patent are unpatentable.

¹ Petitioner’s Motion to Exclude seeks exclusion of Exhibit 2101. As we do not rely on Exhibit 2101, Petitioner’s Motion to Exclude is dismissed as moot. Patent Owner’s Motion to Exclude is discussed below.

B. Real Parties In Interest

Each party identifies itself as the sole real party in interest. Pet. 4; Paper 4, at 1.

C. Related Matters

The parties identify the following related matter involving the '981 patent: *10x Genomics v. Parse Biosciences, Inc.*, 22-cv-01117 (D. Del.). Pet. 4; Paper 4, at 1.

Petitioner also identifies IPR2023-00955 and IPR2023-00958, involving related patents, U.S. Patent No. 10,240,197 B1 (“the '197 patent”) and U.S. Patent No. 10,697,013 B1 (“the '013 patent”). Paper 16, 2.

D. The '981 Patent

The '981 patent is titled “Methods For Analyzing Nucleic Acids From Single Cells.” Ex. 1001, code (54). The focus of the '981 patent Specification, other than the claims, however, is directed to the “reflex method.” For example, the “Summary of the Invention” states in its entirety the following:

Aspects of the present invention are drawn to processes for moving a region of interest in a polynucleotide from a first position to a second position with regard to a domain within the polynucleotide, also referred to as a “reflex method” (or reflex process, reflex sequence process, reflex reaction, and the like). In certain embodiments, the reflex method results in moving a region of interest into functional proximity to specific domain elements present in the polynucleotide (e.g., primer sites and/or MID). Compositions, kits and systems that find use in carrying out the reflex processes described herein are also provided.

Ex. 1001, 1:49–61.

The '981 patent's filing date is August 15, 2017. *Id.*, code (22). However, it claims priority as the sixth in a string of continuation applications and ultimately to a provisional application filed August 20, 2009. Ex. 1001, code (63), 1:7–22. Petitioner does not challenge the asserted priority date of August 20, 2009. *See* Pet. 15 (“[T]he priority date of the '981 Patent is no earlier than August 20, 2009.”).

E. The Challenged Claims

Petitioner challenged all of the claims of the '981 patent, namely claims 1–6. Claim 1 is the sole independent claim and reproduced below:

1. A method of analyzing nucleic acids from a plurality of single cells, the method comprising:
 - (a) providing a sample comprising a plurality of single cells, wherein each single cell of the plurality of single cells comprises a plurality of sample polynucleotides;
 - (b) generating a plurality of tagged polynucleotides from the plurality of sample polynucleotides, wherein each tagged polynucleotide comprises:
 - (i) a sequence from a sample polynucleotide of the plurality of sample polynucleotides; and
 - (ii) a multiplex identifier (MID) sequence comprising:
 - I. a first tag sequence associated with the single cell from which the sample polynucleotide is derived, wherein the first tag sequence is a different sequence for different single cells in the plurality of single cells; and
 - II. a second tag sequence distinguishing the sample polynucleotide from other sample polynucleotides derived from the same single cell;

- (c) sequencing the plurality of tagged polynucleotides to obtain a plurality of identified polynucleotide sequences;
- (d) using the first tag sequence to correlate the identified polynucleotide sequence with the single cell from which the identified polynucleotide sequence is derived; and
- (e) using the second tag sequence to correlate the identified polynucleotide sequence with the sample polynucleotide from which the identified polynucleotide sequence is derived.

Ex. 1001, 30:18–48.

F. The Asserted Grounds of Unpatentability

Petitioner asserts the following grounds of unpatentability:

Claims Challenged	35 U.S.C. § ²	Basis
1–4, 6	103(a)	Linnarsson, McCloskey
5	103(a)	Linnarsson, McCloskey, “further in view of the knowledge of a [person of ordinary skill in the art]”

Pet. 5.

² The Leahy-Smith America Invents Act (“AIA”), Pub. L. No. 112-29, which was enacted September 16, 2011, made amendments to 35 U.S.C. §§ 102 and 103. AIA § 3(b)–(c). Those amendments became effective eighteen months later on March 16, 2013. *Id.* at § 3(n). Because the application from which the ’981 patent issued claims priority to an application filed before March 16, 2013, any citations herein to 35 U.S.C. §§ 102 and 103 are to their pre-AIA versions. *Id.* at § 3(n)(1).

II. ANALYSIS

A. Claim Construction

The challenged claims should be read in light of the specification, as it would be interpreted by one of ordinary skill in the art. *In re Suitco Surface, Inc.*, 603 F.3d 1255, 1260 (Fed. Cir. 2010). Thus, we generally give claim terms their ordinary and customary meaning. *See In re Translogic Tech., Inc.*, 504 F.3d 1249, 1257 (Fed. Cir. 2007) (“The ordinary and customary meaning is the meaning that the term would have to a person of ordinary skill in the art in question.” (internal quotation marks omitted)); *see also* 37 C.F.R. § 42.100(b) (stating that claims are construed in IPRs according to the same standard as used in federal court).

The Petition states that “Petitioner does not believe that any terms require explicit construction in view of the prior art being presented.” Pet. 36. Despite this stated position, Petitioner proposed constructions for “multiplex-identifier (MID)” and “providing a sample.” *Id.* Patent Owner, in its Preliminary Response, stated that it “agrees that at this time, solely for purposes of an institution decision on the Petition, no terms require explicit construction.” Prelim. Resp. 45. In instituting the IPR, we found no need to provide an express construction of any claim term. Inst. Dec. 6.

In the Patent Owner Response, Patent Owner repeats that it “does not believe any term needs construction,” but nonetheless “applies the Petition’s definitions for ‘MID’ and ‘sample.’” PO Resp. 19. The Petition defines *MID* as “‘a tag or combination of tags associated with a polynucleotide whose identity (e.g., the tag DNA sequence) can be used to differentiate polynucleotides in a sample.’” Pet. 36 (quoting Ex. 1001, 6:33–38). The

Petition defines *sample* as ““a quantity of material from a biological, environmental medical, or patient source in which detection, measurement, or labeling of target nucleic acids is sought.”” *Id.* (quoting Ex. 1001, 12:7–10). We apply these same definitions in our analysis.

B. Obviousness Generally

A claim is unpatentable “if the differences between the subject matter sought to be patented and the prior art are such that the subject matter as a whole would have been obvious at the time the invention was made to a person having ordinary skill in the art to which said subject matter pertains.” 35 U.S.C. § 103(a). “Obviousness is a question of law based on underlying facts.” *MobileMedia Ideas LLC v. Apple Inc.*, 780 F.3d 1159, 1167 (Fed. Cir. 2015). The underlying facts include (i) the scope and content of the prior art, (ii) any differences between the prior art and the claimed invention, (iii) the level of ordinary skill in the field of the invention, and (iv) any relevant objective considerations of nonobviousness. *Id.* (citing *Graham v. John Deere of Kan. City*, 383 U.S. 1, 17–18 (1966)). Further, if all the elements of an invention are found in a combination of prior art references, then:

a proper analysis under § 103 requires, inter alia, consideration of two factors: (1) whether the prior art would have suggested to those of ordinary skill in the art that they should make the claimed composition or device, or carry out the claimed process; and (2) whether the prior art would also have revealed that in so making or carrying out, those of ordinary skill would have a reasonable expectation of success.

Medichem, S.A. v. Rolabo, S.L., 437 F.3d 1157, 1164 (Fed. Cir. 2006).

C. Level of Ordinary Skill in the Art

In determining whether an invention would have been obvious at the time it was made, we consider the level of ordinary skill in the pertinent art at the time of the invention. *Graham*, 383 U.S. at 17. “The importance of resolving the level of ordinary skill in the art lies in the necessity of maintaining objectivity in the obviousness inquiry.” *Ryko Mfg. Co. v. Nu-Star, Inc.*, 950 F.2d 714, 718 (Fed. Cir. 1991). The “person having ordinary skill in the art” is a hypothetical construct, from whose vantage point obviousness is assessed. *In re Rouffet*, 149 F.3d 1350, 1357 (Fed. Cir. 1998).

According to Petitioner:

A person of ordinary skill in the art (“POSA”) relating to the ’981 Patent would have had (1) a Masters degree and/or Ph.D. in molecular biology, genetics, chemistry, engineering, or a related discipline and at least two years of post-Masters, postdoctoral, or industry experience or (2) a Bachelor of Science in such disciplines and at least five years of academic or industry experience (including any experimental work toward a graduate degree), relating to DNA sequencing technologies, amplification methods, and gene expression analysis. Further, a POSA would have been familiar with associated tools, methods, and techniques including: (1) polynucleotide tagging; (2) amplification; (3) use of tags to overcome amplification bias; (4) single cell analysis; (5) ligation and primer extension; and (6) sequencing.

Pet. 21–22 (citations omitted).

Patent Owner does not directly dispute Petitioner’s proposal but does offer its own proposal. PO Resp. 18. Patent Owner proposes that the skilled artisan “would have had a Ph.D. in molecular biology, molecular genetics, biochemistry, physics, mechanical engineering, or a related discipline, with

two years of academic, research, or industry experience in the field of genomic or transcriptome sequencing.” *Id.* (citing Ex. 2013 ¶¶25–26).

Both of the parties’ proposals as to the level of ordinary skill in the art are consistent with the disclosure of the ’981 patent and the asserted prior art. The distinction between the parties’ proposals would not change the outcome of this Decision. Thus, we adopt and apply Patent Owner’s proposal.

D. The Prior Art

“A petitioner in an inter partes review may request to cancel as unpatentable 1 or more claims of a patent . . . only on the basis of prior art consisting of patents or printed publications.” 35 U.S.C. § 311(b).

Petitioner here asserts the following two printed publications as prior art:

(1) WO 2010/117620 A2, published Oct. 14, 2010 (“Linnarsson,” Ex. 1003) and (2) McCloskey, Megan L., *et al.*, “Encoding PCR Products with Batch-stamps and Barcodes,” 45 BIOCHEM GENET 761–67 (2007), published Oct. 23, 2007 (“McCloskey,” Ex. 1004).

1. Linnarsson

Linnarsson is titled “Gene Expression Analysis In Single Cells.” Ex. 1003, code (54). Linnarsson discloses a “method for preparing a [complimentary DNA or] cDNA library from a plurality of single cells.” *Id.* at 3:25. In one embodiment

the method includes the steps of releasing [messenger RNA or] mRNA from each single cell to provide a plurality of individual mRNA samples, synthesizing a first strand of cDNA from the mRNA in each individual mRNA sample and incorporating a tag into the cDNA to provide a plurality of tagged cDNA samples,

pooling the tagged cDNA samples and amplifying the pooled cDNA samples to generate a cDNA library having double-stranded cDNA.

Id. at 3:26–30. Linnarsson discloses that the tag may be incorporated during cDNA synthesis via a tagged cDNA synthesis primer (“CDS”) and/or a tagged template switching oligonucleotide (“TSO”). *Id.* at 18:31–32 (“In some aspects of the invention, a tag can be incorporated into the cDNA during its synthesis. For example, the CDS and/or the TSO can include a tag.”).

Linnarsson claims priority to U.S. Provisional Patent Application No. 61/164,759 (“Linnarsson Provisional,” (Ex. 1015)), filed on March 30, 2009. Ex. 1003, code (30). Petitioner argues that the Linnarsson Provisional supports at least one claim of Linnarsson, and, thus, Linnarsson is prior art as of the March 30, 2009, filing date of the Linnarsson Provisional. Pet. 22–24. As such, Petitioner asserts that Linnarsson is prior art to the challenged claims under 35 U.S.C. § 102(e). *See* Pet. 22–23 (“Linnarsson is prior art to the challenged claims even if the Challenged Claims of the ’981 Patent are entitled to their earliest priority date of August 20, 2009.”).

Patent Owner does not dispute that Linnarsson is prior art to the challenged claims. Indeed, Patent Owner points out that “Linnarsson was the basis of multiple rejections by the Office in the application that matured into ’981, as well as ’981’s parent.” PO Resp. 7 (citing Ex. 1001, 2; Ex. 1005, 154–56; Ex. 2004, 1085–87). This is true. In fact, Linnarsson was cited as an anticipatory reference twice during the examination of the ’957 application from which the ’981 patent issued. *See* Ex. 1005, 154 (Dec. 27, 2017, Final Action: “Claim(s) 1–6 are rejected under pre-AIA 35 U.S.C.

102(A)(1)/(A)(2) as being anticipated by Linnarsson.”), 193 (July 10, 2018, Final Action stating the same).

Patent Owner, however, does dispute that Linnarsson is analogous art, which we address below. PO Resp. 68–69.

2. *McCloskey*

McCloskey is titled “Encoding PCR Products with Batch-stamps and Barcodes.” Ex. 1004, 1. McCloskey explains that when employing the polymerase chain reaction (“PCR”), there is “always uncertainty concerning the source of the template DNA that gave rise to a particular PCR product.”

Id. McCloskey reports that it solved this problem by using

an encoding oligonucleotide with five distinct informational regions (Fig. 1): (1) the batch-stamp, unique to each experiment, specifies the DNA source such as the patient or sample identification, and the date of reaction; (2) the random barcode distinguishes among sequences arising from different cell or allele copies; (3) the primer-binding site facilitates specific PCR amplification; (4) the genomic target-binding site targets the locus to be encoded prior to amplification; and (5) the extra 5' bases serve as an internal sentinel of recoding. This encoding oligonucleotide is annealed to denatured DNA and extended by Sequenase (USB) (Fig. 1). Sequenase-extension products are column-filtered to remove any unincorporated, encoding oligonucleotides and amplified using standard PCR protocols (Fig. 1).

Ex. 1004, 2. Such an encoding oligonucleotide with two tag sequences is the heart of what Petitioner relies upon with respect to McCloskey. *See, e.g.,* Pet. 30 (“McCloskey discloses a method of DNA analysis for single-cell samples that utilizes an ‘encoding oligonucleotide’ having ‘five distinct information regions,’ two of which are tag sequences that distinguish, respectively, polynucleotides and samples.”).

According to the face of McCloskey, it was published in 2007, which is more than one year before the earliest possible effective filing date of the challenged claims (i.e., August 20, 2009). Ex. 1004, 1. As such, Petitioner asserts that McCloskey is prior art to the challenged claims under 35 U.S.C. § 102(a) and (b). See Pet. 28–29. Patent Owner disputes that McCloskey is prior art, which we address below. PO Resp. 18.

Patent Owner also disputes that McCloskey is analogous art, which we also address below. *Id.* at 68–69.

3. *Linnarsson and McCloskey are Analogous Art*

“Although § 103 does not, by its terms, define the ‘art to which [the] subject matter [sought to be patented] pertains,’ this determination is frequently couched in terms of whether the art is analogous or not, i.e., whether the art is ‘too remote to be treated as prior art.’” *In re Clay*, 966 F.2d 656, 658 (Fed. Cir. 1992) (quoting *In re Sovish*, 769 F.2d 738, 741 (Fed. Cir. 1985)). “Whether a reference in the prior art is ‘analogous’ is a fact question.” *Clay*, 966 F.2d at 658. “Two criteria have evolved for determining whether prior art is analogous: (1) whether the art is from the same field of endeavor, regardless of the problem addressed, and (2) if the reference is not within the field of the inventor’s endeavor, whether the reference still is reasonably pertinent to the particular problem with which the inventor is involved.” *Id.* at 658–59.

Patent Owner argues that “Nowhere does Petitioner assert—let alone attempt to show—that either of the two references named in its grounds (primary reference Linnarsson and secondary reference McCloskey) is analogous art *to ’981*.” PO Resp. 68 (citing Pet. 52–53). As Patent Owner

explains it, the Petition asserts that Linnarsson and McCloskey are analogous *to each other* but does not assert that they are analogous *to the challenged claims*. *Id.* (citing *Sanofi-Aventis Deutschland GmbH v. Mylan Pharms. Inc.*, 66 F.4th 1373, 1380 (Fed. Cir. 2023)).

As we stated in the Institution Decision

We agree with Patent Owner that *Sanofi-Aventis* holds that a petitioner bears the burden of persuasion to establish that an asserted prior art reference is analogous art. . . .

However, we disagree with Patent Owner that the Petition must be denied for failing to state that the cited prior art references are analogous art “to the ’981 patent.” Rather, as *Sanofi-Aventis* further stated: “A petitioner is not required to anticipate and raise analogous art arguments in its petition; instead a petitioner can use its reply to ‘respond to arguments raised in the corresponding opposition, patent owner preliminary response, patent owner response, or decision on institution.’” *Sanofi-Aventis*, 66 F.4th at 1379 (quoting 37 C.F.R. § 42.23). Accordingly, we need not address this issue for purposes of deciding whether to institute trial.

Inst. Dec. 18.

In its Petition, Petitioner asserted that “Linnarsson and McCloskey are analogous art that focus on significantly the same subject matter” and “are thus in the same field of endeavor.” Pet. 52. Patent Owner interprets the phrases “same subject matter” and “same field of endeavor” as meaning the *same subject matter as each other* and the *same field of endeavor as each other*. Thus, according to Patent Owner, Petitioner did not make the requisite assertion. This is a reasonable interpretation. This is not the end of the analysis, however.

In its Reply, Petitioner argues that Linnarsson is analogous art because both it and the ’981 patent “deal with ‘analysis of gene expression

in single cells or in a plurality of single cells’ based on nucleic acid tagging and sequencing.” Pet. Reply 25 (citing Ex. 1003, Abstr., 1:6-9, 3:25–4:2, 9:23–10:2, 10:15–30, Fig. 11, claims 1, 26, 27; Ex. 1001, code (54), Abstr., 6:33–7:22, claim 1; E. 1059 ¶¶125–27). In its Reply, Petitioner also argues that McCloskey is analogous art because both it and the ’981 patent “deal with tagging polynucleotides for amplification,[] with McCloskey even teaching a two-part tag that identifies the ‘DNA source such as the patient or sample identification’ and also ‘distinguishes among sequences arising from different cell or allele copies.’” Pet. Reply 25 (citing Ex. 1004, 2, Fig. 1; Ex. 1001, code (54), Abstr., 6:33–7:22, claim 1; Ex. 1059 ¶¶128–30).

We find that both references are analogous art to the ’981 patent, as there is ample supporting evidence in the record, including that to which the Reply cites (citations identified in the immediately preceding paragraph). Indeed, Patent Owner does not dispute the merits of either argument. PO Sur-Reply 27–28. Rather, Patent Owner, argues that we “should not now permit Petitioner to rewrite its clearly-deficient Petition.” *Id.* at 28.

Petitioner’s Reply does not re-write its Petition. What Petitioner’s Reply does, in response to Patent Owner’s arguments, is precisely what *Sanofi-Aventis* contemplates as being proper in a reply brief. Indeed, in ruling against the petitioner in *Sanofi-Aventis*, the Federal Circuit specifically noted that the petitioner could have used its reply brief “to explain how [the prior art reference] is analogous to the [challenged] patent” *but did not do so*. 66 F.4th at 1379–1380. Here, in contrast, Petitioner used its Reply brief to explain how Linnarsson and McCloskey are analogous to the ’981 patent. Petitioner’s explanation in its Reply brief was both proper and persuasive.

4. *McCloskey is a Prior Art Printed Publication; Patent Owner's Motion to Exclude is Granted-in-Part and Denied-in-Part*

Section 311(b) provides: “A petitioner in an inter partes review may request to cancel as unpatentable 1 or more claims of a patent only on a ground that could be raised under section 102 or 103 and *only on the basis of prior art consisting of patents or printed publication.*” 35 U.S.C. § 311(b) (emphasis added).

McCloskey is an article titled “Encoding PCR Products with Batch-stamps and Barcodes,” and authored by Megan L. McCloskey, Reinhard Stoger, R. Scott Hansen, and Charles D. Laird. Ex. 1004, 1. Petitioner argues that McCloskey is a prior art printed publication because it was published in BIOCHEMICAL GENETICS in 2007, as indicated on the cover of McCloskey. Pet. 29 (citing Ex. 1004, 1; Ex. 1028). Exhibit 1028 does not support Petitioner’s assertion regarding McCloskey and appears to have been prepared for another matter. See Pet. Reply 3 n.2 (“The Petition inadvertently included an incorrect declaration from Dr. Hall-Ellis. See EX1028.”).

Patent Owner argues that “Petitioner failed to establish McCloskey is, in fact, prior art to ’981.” PO Resp. 18 (citing *Standard Innovation Corp. v. Lelo, Inc.*, No. IPR2014-00148, Paper 41 at 18–28 (Apr. 23, 2015); *Dropbox, Inc. v. Synchronoss Techs.*, No. IPR2016-00850, Paper 41 at 18–23 (Oct. 5, 2016)). Patent Owner does not explain how Petitioner allegedly failed to prove McCloskey is prior art. Instead, Patent Owner implies, via its description of *Standard Innovation’s* holding, that Petitioner

failed to prove that McCloskey is what it has been represented to be or that it was actually published in 2007. PO Resp. 18.

With its Reply, Petitioner submitted a second declaration from the same witness who executed the inadvertently included Exhibit 1028, i.e., Sylvia Hall-Ellis, Ph.D. Ex. 1065. In her second declaration, Dr. Hall-Ellis includes an Exhibit A, which she testifies includes a copy of McCloskey that she obtained from Linda Hall Library of Science, Engineering & Technology. *Id.* ¶¶43–50, “Exhibit A.” Dr. Hall-Ellis testifies that McCloskey is the article it appears to be, was published in BIOCHEMICAL GENETICS in 2007, and was received by Linda Hall Library by January 3, 2008, at the latest. *Id.* ¶¶43–50.

Patent Owner filed a Motion to Exclude the second Hall-Ellis declaration (Ex. 1065) and McCloskey (Ex. 1004). PO Mot. To Exclude.

With respect to the second Hall-Ellis declaration, Patent Owner objected to its admissibility on the basis that, among other things, it “should have been served, but was not, as supplemental evidence within ten business days of [Patent Owner] 10x’s timely objection under 37 C.F.R. § 42.64(b)(2).” Paper 23, 3. Relatedly, Patent Owner had previously objected -- on October 27, 2023 -- to McCloskey as allegedly unauthenticated, undated (absent hearsay), and thus irrelevant as not a prior art printed publication as to the challenged claims. Paper 11, 1–2.

Clearly, the second Hall-Ellis declaration is offered to prove that McCloskey is a prior art printed publication. *See, e.g.*, Ex. 1065 ¶50 (“In view of the foregoing, it is my opinion that [McCloskey] was publicly available no later than [January 3, 2008].”). Thus, the second Hall-Ellis declaration should have been served within ten business days of Patent

Owner's October, 27, 2023, objection to the admissibility of McCloskey. See 37 C.F.R. § 42.64(b)(2) ("The party relying on evidence to which an objection is timely served may respond to the objection by serving supplemental evidence within ten business days of service of the objection."). Because it was not filed until March 29, 2024, Patent Owner's Motion to Exclude is granted with respect to the second Hall-Ellis declaration.

In its Opposition to the Motion to Exclude, Petitioner argues that, under *Hulu LLC v. Sound View Innovations, LLC*, No. IPR2018-01039, Paper 29 (Dec. 20, 2019) (precedential), it was allowed to wait until its Reply to file the second Hall-Ellis declaration. Paper 37, 1–2. The Board in *Hulu* does state that, "if the patent owner challenges a reference's status as a printed publication, a petitioner may submit a supporting declaration with its reply to further support its argument that a reference qualifies as a printed publication." *Hulu*, at 15 (37 C.F.R. § 42.23). However, the Board's statement was referring to the scope of permissible reply arguments under 37 C.F.R. § 42.23(b). See *Hulu* at 14 (quoting 37 C.F.R. § 42.23(b)'s requirement that "[a] reply may only respond to arguments raised in the corresponding opposition, patent owner preliminary response, or patent owner response"). The Board in *Hulu* does not purport to interpret 37 C.F.R. § 42.64 or even discuss supplemental evidence. Petitioner clearly missed the deadline imposed by 37 C.F.R. § 42.64(b)(2), which was triggered by Patent Owner's October 27, 2023, objection to McCloskey. Patent Owner's Motion to Exclude is granted with respect to the second Hall-Ellis declaration (i.e., Ex. 1065).

Despite the exclusion of the second Hall-Ellis declaration, there is ample evidence in the record demonstrating that McCloskey is a prior art printed publication. McCloskey itself states: “Received: 31 January 2007 / Accepted: 29 May 2007 / Published online: 23 October 2007.” Ex. 1004, 1. McCloskey additionally states that it was published in a 2007 print issue of “Biochem Genet,” with Petitioner adding that the full name of the journal is BIOCHEMICAL GENETICS and that McCloskey was published in a December 2007 volume of the journal. *Id.*; Pet. 29. McCloskey also includes a copyright notice by “Springer Science+Business Media, LLC” with a date of date of 2007. Ex. 1004, 1. Patent Owner argues that this is unreliable hearsay that should be excluded. PO Mot. To Exclude 4 (citing *Neste Oil Oyj v. REG Synthetic Fuels, LLC*, No. IPR2013-00578, Paper 53 at 5–7 (Mar. 12, 2015); *Standard Innovation Corp. v. Lelo, Inc.*, No. IPR2014-00148, Paper 42³ at 13–18, 21–22 (Apr. 23, 2015)). We are not persuaded by Patent Owner’s argument or its reliance on these cases. The facts before us are different.

In *Neste Oil*, the Board “note[d] that the alleged ‘stamps’ of Exhibits 1125 and 1132 are actually handwritten dates, added to the documents by an unknown person at an unknown time.” *Neste Oil*, Paper 53 at 6. In contrast, the dates relied upon by Petitioner are printed, appear to have been included at the time of printing, and appear to have been printed by, or at the direction

³ Patent Owner actually cited to “Pap.41” of *Standard Innovation*. PO Mot. To Exclude 4. There is no Paper 41 in the record of that case. We presume Patent Owner intended to cite Paper 42, which is the final written decision in that case.

of, the journal and/or copyright holder, both of which are identified by McCloskey. Ex. 1004, 1.

In *Standard Innovation*, the alleged prior art exhibits were “printouts of ... webpages” bearing a copyright indication of “© 2003-2013,” which the petitioner argued was proof of subset of the webpages being disseminated publicly in 2003. *Standard Innovation*, Paper 42 at 13. This is a far cry from McCloskey, which is a journal article and which includes specific date information, not only of publishing, but also of the dates the journal received and accepted the article for publication as well as the date of copyright. Ex. 1004, 1.

That McCloskey is what it purports to be (a journal article) and that it was published in 2007 are corroborated by multiple pieces of evidence.

McCloskey names four authors, all four of whom are named inventors of US 2007/0020640 A1, published Jan. 25, 2007 (“McCloskey ’640”). Ex. 1004, 1, Ex. 2001, code (76). Patent Owner concedes that McCloskey and McCloskey ’640 involve the same subject matter, having argued that the consideration of McCloskey ’640 supported discretionary denial of the Petition based on McCloskey. *See, e.g.*, Prelim. Resp. 22 (“Petitioner’s secondary reference, an article by McCloskey, is at best cumulative of the far more detailed McCloskey ’640, which ***was raised by the Office and substantively discussed as the basis for a potential rejection using its asserted disclosure of a polynucleotide-specific tag—the same purported teaching for which Petitioner relies on McCloskey here.***”). Notably, McCloskey ’640 was published on January 25, 2007. Ex. 2001, code (43). That is just days before McCloskey was received by BIOCHEMICAL GENETICS on January 31, 2007. *See* Ex. 1004, 1 (“Received:

31 January 2007 / Accepted: 29 May 2007 / Published online: 23 October 2007”). In other words, it appears that the authors sought to have their research published in a scientific journal promptly after the subject matter of their research had been published by the Office.

Further, the existence of McCloskey and its publication year of 2007 is confirmed by other sources in the record. See Ex. 1029, code (56) (United States Patent identifying as a cited reference: “McCloskey, et al. Encoding Pcr products with batch-stamps and barcodes. Biochem Genet. Dec. 2007;45(11-12):761-7. Epub Oct. 23, 2007.”); Ex. 1053, code (56) (same); Ex. 1060 (article’s bibliography citing “McCloskey ML, Stöger R, Hansen RS, Laird CD. Encoding PCR products with batch-stamps and barcodes. Biochem. Genet 2007;45:761–767. [PubMed: 17955361]”).

Patent Owner’s Motion to Exclude is denied with respect to McCloskey (i.e., Ex. 1004).

In sum, Petitioner has shown that McCloskey is a printed publication as of 2007 and, thus, prior art to the challenged claims of the ’981 patent.

III. Ground I (Claims 1–4 and 6)

A. Claim 1

Petitioner argues that claim 1 would have been obvious over Linnarsson and McCloskey. Pet. 37–59.

In mapping the prior art to claim 1, the Petition breaks the claim into nine portions and organizes its arguments accordingly. See Pet. 37–59 (addressing limitations labeled by Petitioner as [1A] through [1I]). We largely follow suit.

1. *“A method of analyzing nucleic acids from a plurality of single cells, the method comprising”*

Neither party argues that the preamble is a limitation. Nonetheless, Petitioner demonstrates that Linnarsson teaches a method of analyzing nucleic acids from a plurality of single cells. Pet. 37. Indeed, as pointed out by Petitioner, Linnarsson teaches “methods and compositions for the analysis of gene expression in single cells or in a plurality of single cells.” Ex. 1003, 9:23–25 (quoted at Pet. 37). As further pointed by Petitioner, Linnarsson teaches one embodiment involving the following steps:

(A) the tissue of interest is dissected; (B) a plurality of single cells are selected; (C) single cells are placed in separate wells of a 96-well plate and lysed; tagged reverse transcription is performed on each sample to produce cDNA; (D) cDNA samples are pooled and amplified; (E) sequencing is performed to obtain 100 million reads; and (F) identification of expressed genes and identification of cells from which they originated.

Ex. 1003, 4:8–13 (quoted at Pet 37). Patent Owner does not dispute that Linnarsson teaches the preamble of claim 1. *See generally* PO Resp. We are persuaded that it does.

2. *“providing a sample comprising a plurality of single cells, wherein each single cell of the plurality of single cells comprises a plurality of sample polynucleotides”*

For this limitation, Petitioner relies on Linnarsson alone. *See* Pet. 38 (“Linnarsson discloses this claim element.”). Patent Owner does not dispute that Linnarsson teaches this limitation. *See generally* PO Resp. We are persuaded that it does in view of what we have already block-quoted above in our discussion of the preamble, as well as additional excerpts cited by

Petitioner. See Pet. 38–39 (citing Ex. 1003, 3:26–30, 4:8–11, 13:11–12, 13:28, 22:22–23).

3. “*generating a plurality of tagged polynucleotides from the plurality of sample polynucleotides*”

For this limitation, Petitioner relies on Linnarsson alone. See Pet. 39 (“Linnarsson discloses this claim element.”). Patent Owner does not dispute that Linnarsson teaches this limitation. See *generally* PO Resp. We are persuaded that it does in view of what we have already block-quoted above in our discussion of the preamble, as well as additional excerpts cited by Petitioner.

For example, Petitioner quotes Linnarsson’s disclosure that, “[a]s the tag is present in the CDS and/or the TSO it will be incorporated into the cDNA during its synthesis and can therefore act as a ‘barcode’ to identify the cDNA.” Ex. 1003, 19:4–7 (quoted at Pet. 40–41). Petitioner also relies on various figures of Linnarsson, including Figures 3, 4, and 11. Pet. 41–43.

Figure 3 of Linnarsson is reproduced below.

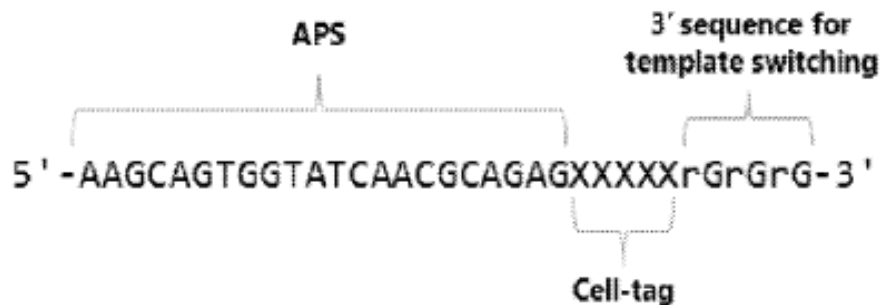


Figure 3 of Linnarsson above “shows an example of a template switching oligonucleotide [or TSO] comprising a 5' amplification primer sequence (APS), a *cell tag* and a 3' sequence for template switching.” Ex. 1003, 4:15–16 (emphasis added).

that Linnarsson teaches this limitation. *See generally* PO Resp. We are persuaded that it does in view of what we have already block-quoted above in our discussion of the preamble (i.e., “tagged reverse transcription is performed on each sample to produce cDNA” and “cDNA samples are pooled and amplified”) as well as Figure 2 as explained by Dr. Cooper.

Figure 2 “shows the synthesis of cDNA by template switching.” Ex. 1003, 4:14. Dr. Cooper testifies in connection with Figure 2 that a person of ordinary skill in the art “would have understood that one strand of the resulting double-stranded cDNA would be identical to the sequence of the sample mRNA because a cDNA synthesized from a sample mRNA comprises the same sample sequence,” except that the cDNA would employ thymine wherever mRNA employed uracil. Ex. 1002 ¶152. As for the noted exception, Dr. Cooper testifies that “[t]he presence of thymine in the cDNA indicates the presence of uracil in the mRNA, thus preserving the sequence information,” and “convey[ing] the same genetic information.” *Id.* Patent Owner does not dispute this testimony.

Linnarsson clearly teaches this limitation.

5. *“wherein each tagged polynucleotide comprises: . . .
(ii) a multiplex identifier (MID) sequence comprising:
I. a first tag sequence associated with the single cell
from which the sample polynucleotide is derived,
wherein the first tag sequence is a different sequence
for different single cells in the plurality of single cells;
and II. a second tag sequence distinguishing the sample
polynucleotide from other sample polynucleotides
derived from the same single cell”*

For brevity, we will refer to the above-quoted claim language, which the Petition breaks into limitations it labels [1E] and [1F], as the “dual-tag MID” limitation.

This dual-tag MID limitation is the basis on which the '981 claims were distinguished over Linnarsson during prosecution of the '981 patent. *See* Ex. 1005, 215 (Patent Owner: “Linnarsson does not describe the ‘second tag sequence’ element in the claims.”), 238 (Examiner: “The closest prior art, Linnarsson (WO 2010/117620), does not teach or suggest generating a plurality of tagged polynucleotides comprising an identifier sequence of instant claim 1(ii), [which] comprises a first tag sequence, which distinguishes each single cell in a sample of a plurality of cells, and a second tag sequence, which distinguishes a sample polynucleotide from other sample polynucleotides derived from a single cell.”).

To address the gap between Linnarsson and claim 1, Petitioner turns to McCloskey. Pet. 50. Petitioner persuasively argues that McCloskey teaches a MID with two tag sequences. *See* Pet. 51 (“As shown [in Ex. 1004, Fig. 1A], McCloskey’s encoding oligonucleotide comprises two tag sequences: (1) a ‘batch-stamp’ sequence . . . to identify the experimental batch or genomic source (e.g., a single cell) and (2) a ‘barcode’ . . . to

distinguish polynucleotide sequences “arising from different cell or allele copies.”). As noted by Petitioner, in McCloskey, “the batch-stamp, unique to each experiment, specifies the DNA source such as the patient or sample identification, and the date of reaction” and “the random barcode distinguishes among sequences arising from different cell or allele copies.” Pet. 51 (quoting Ex. 1004, 2).

Figure 1A of McCloskey is reproduced below.

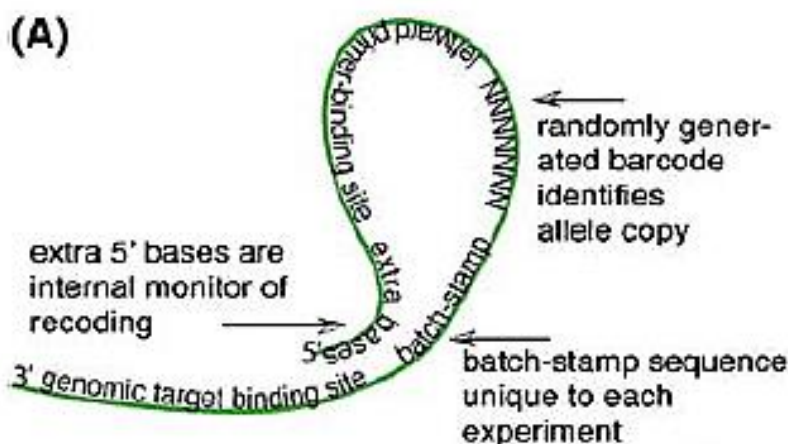


Figure 1A of McCloskey above shows encoding oligonucleotide having “five distinct regions,” namely “a genomic target-binding site, a batch-stamp, a barcode, a primer-binding site for PCR, and extra bases that serve as internal sentinels of recoding as well as tethering the oligonucleotide to itself.” Ex. 1004, 3. Thus, McCloskey clearly teaches a dual-tag MID.

The Petition additionally relies on McCloskey for teaching “that the ‘batch-stamps and barcodes [should] be used when amplifying irreplaceable DNAs *and* cDNAs for . . . *single cell* . . . analyses,’ and that its method of tagging applies ‘to single-stranded [DNA or] cDNA templates.’” Pet. 51

(quoting Ex. 1004, 1–2).⁴ Based on these teachings and declaration testimony of Gregor Cooper, Ph.D., Petitioner persuasively argues that a person of ordinary skill in the art would “would have understood that McCloskey’s reference to ‘different cell or allele copies’ encompasses cDNA (i.e., sample polynucleotides) from single cell experiments.” *Id.* at 51 (citing Ex. 1002 ¶¶65, 184; Ex. 1004, 1–2).

Ultimately, Petitioner argues that “[i]t would have been obvious to combine Linnarsson’s methods of cDNA library preparation and analyzing tagged nucleic acids with McCloskey’s teachings to use a MID (i.e., encoding oligonucleotide) comprising a second tag sequence that distinguishes sample polynucleotides derived from the same single cell.” Pet. 52. That modification, according to Petitioner, would have been to “conceptually divide each of Linnarsson’s cell-tag sequence into two portions, wherein the first portion performs the function of McCloskey’s ‘batch-stamp,’ used to track individual cells, and the second portion performs the function of McCloskey’s ‘barcode,’ used to differentiate polynucleotides within a cell.” Ex. 1002 ¶192.

We are persuaded that such a modification would have resulted in the claimed subject matter, including the use of a dual-tag MID. Further, and as discussed below, we are persuaded that a person of ordinary skill in the art would have had a sufficient reason to make the modification and a reasonable expectation of success in doing so.

⁴ The Petition cites to pages 1 and 5 of Exhibit 1004 for these quotes. Pet. 51. However, we located the quotes on pages 1 and 2.

6. “*sequencing the plurality of tagged polynucleotides to obtain a plurality of identified polynucleotide sequences*”

For this limitation, Petitioner relies on Linnarsson alone. See Pet. 56 (“Linnarsson discloses this claim element.”). Patent Owner argues that Petitioner has not met its burden to show that Linnarsson teaches this limitation. PO Resp. 22–26. We turn to Petitioner’s arguments first, followed by those of Patent Owner.

Petitioner argues that “Linnarsson teaches that after generating the tagged polynucleotides, ‘sequencing is performed’ thereon.” Pet. 57 (quoting Ex. 1003, 4:12). Petitioner directs us to additional portions of Linnarsson that discuss sequencing, including its teaching that the generated “cDNA library can be sequenced to provide an analysis of gene expression in single cells or in a plurality of single cells.” Ex. 1003, 21:24–25 (cited at Pet. 57); *see also* Ex. 1003, 19:28–32, 21:13–25 (cited at Pet. 57).

Patent Owner argues, however, that Petitioner fails to meet its burden as to the “sequencing” limitation. This is so, according to Patent Owner, because Petitioner failed to show that Linnarsson discloses an unreasonably narrow construction of the limitation, which Patent Owner itself disavows. Patent Owner’s position, which we explain in more detail below, is a curious one. It is not persuasive.

In the Patent Owner Response, Patent Owner argued that Petitioner has not shown that Linnarsson teaches sequencing the *entirety* of cDNA molecules and has shown only that Linnarsson teaches sequencing of fragments of the cDNA molecules. See PO Resp. 23 (“Linnarsson repeatedly discloses that the tagged sequences are fragmented, and that only

the 5' fragment is sequenced.”), 24 (“Petitioner’s own citations fail to show Linnarsson discloses sequencing the entirety of the ‘tagged polynucleotide.’”). Petitioner replied that Patent Owner’s argument was based on a new construction and one that it notably “did not seek it in parallel district court proceedings.” Pet. Reply 3–4 (citing Ex. 1058⁵). In its Sur-Reply and at the hearing, Patent Owner explicitly disavowed any construction that would require sequencing tagged polynucleotides in their entirety. *See* PO Sur-Reply 8–9 (Patent Owner stating that its prior arguments are “[f]ar from construing the claims to require sequencing the **entirety** of the tagged polynucleotides.” (brackets and internal quotation marks omitted)); Tr. 50:1–4 (Counsel for Patent Owner: “With respect to the scope of the claims and sequencing -- the sequencing limitation, we are not asking for construction that the claims require sequencing the entire polynucleotide. We aren’t construing the claims to require that the entire polynucleotide is sequenced.”).

Nevertheless, Patent Owner maintains that Petitioner failed to prove that Linnarsson teaches the “sequencing” limitation. According to Patent Owner, (1) Petitioner originally took the position that Linnarsson satisfies the “sequencing” limitation by virtue of it allegedly sequencing the entire lengths of tagged polynucleotides; (2) Linnarsson does actually sequence the entire lengths of tagged polynucleotides; and (3) Petitioner, in its Reply, pivoted to arguing that Linnarsson satisfies the “sequencing” limitation by

⁵ Exhibit 1058 is a Joint Claim Construction Brief filed December 7, 2023, in the related lawsuit, *10x Genomics v. Parse Biosciences, Inc.*, 22-cv-01117 (D. Del.).

virtue of its sequencing fragments of tagged polynucleotides. PO Sur-Reply 8–9. Patent Owner argues that “Petitioner’s new reply theory should be rejected.” *Id.* at 9. We disagree with Patent Owner’s characterization of Petitioner’s arguments.

Petitioner never argued that the “sequencing” limitation requires sequencing tagged polynucleotides in their entirety. Nor did Petitioner assert that Linnarsson sequences tagged and unfragmented cDNA molecules in their entirety. In any event, the evidence cited by Petitioner, *in its Petition*, that Linnarsson teaches “sequencing the plurality of tagged polynucleotides to obtain a plurality of identified polynucleotide sequences,” as recited in claim 1, is robust. Pet. 57 (citing Ex. 1003, 4:8–13, 19:28–32, 21:13–25). We are persuaded that Linnarsson teaches this limitation.

7. *“using the first tag sequence to correlate the identified polynucleotide sequence with the single cell from which the identified polynucleotide sequence is derived; and
[]using the second tag sequence to correlate the identified polynucleotide sequence with the sample polynucleotide from which the identified polynucleotide sequence is derived.”*

For brevity, we will refer to the above-quoted claim language, which the Petition breaks into limitations it labels [1H] and [1I], as the “correlate” limitations.

Petitioner argues that the “correlate” limitations are satisfied by the proposed modification of Linnarsson in view of McCloskey. Pet. 57–59. In explaining its case, Petitioner starts by arguing that Linnarsson alone satisfies the “first tag ... correlate” limitation because Linnarsson’s native tag tracks a polynucleotide to its source *cell*. Pet. 57 (citing Ex. 1003,

19:5–11). Petitioner then proposes a modification in which Linnarsson’s native tag would be “conceptually divide[d] . . . into two portions, wherein the first portion performs the function of McCloskey’s ‘batch-stamp,’ used to track individual cells, and the second portion performs the function of McCloskey’s ‘barcode,’ used to differentiate polynucleotides within a cell.” Ex. 1002 ¶192 (cited at Pet. 56).

Patent Owner argues that Petitioner inconsistently maps the first and second tags in the proposed modification to what they must be used “to correlate,” as recited in claim 1. PO Resp. 26. Specifically, Patent Owner points out that “Petitioner maps the ‘first tag sequence’ . . . to Linnarsson’s ‘cell-tag,’” but then “abandon[s] Linnarsson’s relied-on existing cell-tag . . . and rel[ies] on a modified version that functions like McCloskey’s batch-stamp as the ‘first tag sequence.’” *Id.* at 26–27. Patent Owner characterizes Petitioner’s modification as an attempt to “rely on disclosure in one reference to cover a limitation, and then *reverse course* and rely on disclosure from *another* reference to modify that same limitation when it appears later.” *Id.* at 28. This argument is not persuasive. Petitioner is not reversing course from reliance on Linnarsson’s native tag. Rather, its proposal is to *modify* the native Linnarsson tag “into two portions, wherein the first portion performs the function of McCloskey’s ‘batch-stamp,’ used to track individual cells, and the second portion performs the function of McCloskey’s ‘barcode,’ used to differentiate polynucleotides within a cell.” Ex. 1002 ¶192.

Patent Owner also argues that “Petitioner never explains how McCloskey’s batch-stamp disclosure actually functions as a cell tag.” PO Resp. 28. This is fatal to Petitioner’s case, according to Patent Owner,

because McCloskey’s “batch-stamp distinguishes between nucleic acids arising from *all the cells in the sample (batch)*, on one hand, and the contaminant nucleic acids arising from cells *from different, prior samples (other batches with different batch-stamps) or other contaminant nucleic acids (with no batch-stamp)*, on the other hand.” *Id.* (citing Ex. 1004, 3–4; Ex. 1002 ¶¶99, 103). This argument is not persuasive for a couple of reasons.

First, Patent Owner’s factual premise regarding McCloskey is incorrect. Nothing in McCloskey precludes a “sample” from being a single cell. In fact, as Petitioner notes in its Reply, McCloskey teaches that its methods can be used for “single cell” analyses. Pet. Reply 9–10; Ex. 1004, 7 (“We recommend that batch-stamps and barcodes be used when amplifying irreplaceable DNAs and cDNAs for forensic, clinical, single cell, and ancient DNA analyses.”).

Second, and more importantly, Petitioner does not rely on McCloskey for teaching a cell-specific tag. Petitioner cites Linnarsson for that feature, and Linnarsson does teach it. Pet. 47–49; *see, e.g.*, Ex. 1003, Fig. 3 (labeling “Cell-tag”), 19:9–10 (“the tag can be used to identify which single cell from each cDNA sample originated”).

We are persuaded that the proposed modification of Linnarsson in view of McCloskey satisfies both “correlate” limitations.

8. Reason To Combine

“[A] patent composed of several elements is not proved obvious merely by demonstrating that each of its elements was, independently, known in the prior art.” *KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 418

(2007). “[I]t can be important to identify a reason that would have prompted a person of ordinary skill in the relevant field to combine the elements in the way the claimed new invention does.” *Id.*; see also *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006) (“[R]ejections on obviousness grounds cannot be sustained by mere conclusory statements; instead, there must be some articulated reasoning with some rational underpinning to support the legal conclusion of obviousness”).

Petitioner argues that “[i]t would have been obvious to combine Linnarsson’s methods of cDNA library preparation and analyzing tagged nucleic acids with McCloskey’s teachings to use a MID (*i.e.*, encoding oligonucleotide) comprising a second tag sequence that distinguishes sample polynucleotides derived from the same single cell” and presents two rationales in support. Pet. 52. First, Petitioner argues that doing so would have been an “obvious design choice.” *Id.* Second, Petitioner argues a skilled artisan would have made the combination because Linnarsson discloses a problem, PCR-induced amplification bias, for which McCloskey teaches a solution. *Id.* at 53. As discussed below, we find each of these rationales independently sufficient to support the proposed modification.

a) Obvious Design Choice

The Supreme Court has “set forth an expansive and flexible approach” to “the question of obviousness” under 35 U.S.C. § 103. *KSR*, 550 U.S. at 415. In *KSR*, the Supreme Court held the following:

When a work is available in one field of endeavor, design incentives and other market forces can prompt variations of it, either in the same field or a different one. If a person of ordinary skill can implement a predictable variation, § 103 likely bars its patentability. For the same reason, if a technique has been used

to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill. [The relevant question is] whether the improvement is more than the predictable use of prior art elements according to their established functions.

KSR, 550 U.S. at 417; *see also Uber Techs., Inc. v. X One, Inc.*, 957 F.3d 1334, 1338 (Fed. Cir. 2020) (quoting the same excerpt from *KSR* and referring to it as a “discussion of design choices and predictable variations”).

Here, Petitioner argues that “using a MID comprising a second tag to distinguish sample polynucleotides derived from the same single cell [in Linnarsson] would have been an obvious design choice” in view of McCloskey’s teaching and the fact that both references relate to tagging and amplifying polynucleotides for genetic analysis. Pet. 52 (citing Ex. 1002 ¶¶187, 192). Indeed, Linnarsson teaches the use of an encoding oligonucleotide with one tag sequence for correlating tagged polynucleotides according to one feature (cell origin), and McCloskey teaches the use of an encoding oligonucleotide with two tag sequences for correlating tagged polynucleotides according to two features (sample origin and polynucleotide). Ex. 1003, 4; Ex. 1004, 2.

Patent Owner does not show that the proposed modification would have been beyond the level of skill in the art. Nor does Patent Owner show that the modification would have resulted in anything beyond what would have been predicted based on these prior art teachings. In fact, the Patent Owner Response does not even address Petitioner’s design choice rationale. PO Resp. 33–40.

We find Petitioner’s un rebutted rationale of obvious design choice persuasive.

b) Solving Amplification Bias

Although we find Petitioner’s “design choice” rationale adequate, we also address Petitioner’s additional rationale that a skilled artisan would have made the combination because Linnarsson discloses a problem, PCR-induced amplification bias, for which McCloskey teaches a solution. Pet. 53 (citing Ex. 1002 ¶¶188–191; Ex. 1003, 2:3–15, 11:6–10; Ex. 1004, Abstract). Indeed, Linnarsson explicitly states “there is potential amplification bias” in its described methods. Ex. 1003, 2:12–13; *see also id.* at 11:6–10 (describing a way to reduce the amount of amplification needed and thus also reduce the amount amplification bias). And, McCloskey teaches that its method of using batch-stamps and barcodes “solve[s] th[e] source-uncertainty problem for DNA sequences generated by standard PCR,” which can be caused by “contamination, biased amplification, and product redundancy.” Ex. 1004, Abstr.

Patent Owner argues that the amplification bias rationale “[f]ails” because the proposed modification would not actually solve the problem of amplification bias. PO Resp. 33. This is so, Patent Owner explains, because “McCloskey does not have enough barcodes to tag each mRNA molecule of Linnarsson.” *Id.* (emphasis omitted). Thus, according to Patent Owner, it “would be impossible, using McCloskey’s 7-nucleotide barcode (amounting to only 16,348 unique barcodes), to distinguish the 50,000–300,000 mRNA molecules Petitioner admits POSITA would have expected to be present in the cell.” *Id.* at 34.

Petitioner replies that (1) Linnarsson is not limited to mammalian cells, (2) neither the claims of the ’981 patent nor Petitioner’s rationale requires *all* of the mRNAs present in a cell to be tagged, and (3)

McCloskey's disclosure is not limited to 16,384 unique barcodes. Pet. Reply. 11–13. We agree with Petitioner on all accounts as discussed below.

i. Linnarsson is not Limited to Mammalian Cells

Petitioner argues that neither the claims of the '981 patent nor the disclosure of Linnarsson is limited to mammalian cells. Pet. Reply 11. Petitioner quotes Linnarsson's disclosure that, "in general, cells from any population can be used in the methods, such as a population of prokaryotic or eukaryotic single celled organisms including bacteria or yeast." *Id.* (quoting Ex. 1003, 13:14–16). Petitioner also cites to Dr. Cooper's testimony regarding various bacteria that are known to contain less than 5000 different mRNA molecules. *Id.* at 11–12 (citing Ex. 1059 ¶¶86–87).

Patent Owner argues that "the Petition repeatedly relied on eukaryotic cells" from Linnarsson and thus it is improper for the Petitioner, in its Reply, to rely on yeast or bacteria.⁶ PO Sur-Reply 2 (citing Pet. 37–39, 41–48). We disagree. The Petition's discussion of the cells in Linnarsson was not limited to mammals or even to eukaryotes. Instead, in multiple instances, the Petition discusses the cells in Linnarsson generically without limiting them to any particular kingdom (e.g., animals, yeast, bacteria) or class (e.g. mammals). *See, e.g.*, Pet. 37 ("Linnarsson discloses a method of analyzing nucleic acids from a plurality of single cells: the 'present invention provides methods and compositions for the analysis of gene expression in single cells or in a plurality of single cells.'" (quoting Ex. 1003, 9:23–25)), 38

⁶ Patent Owner's argument is not factually persuasive. Also, it is not logical given that yeast is eukaryotic. *See* Ex. 1057, 131:6–7 (Patent Owner's declarant, John Quackenbush, Ph.D., testifying on cross-examination that yeast is eukaryotic).

(“Linnarsson explains that the sample of single cells can take various forms.” (citing Ex. 1003, 13:11–12)). In fact, we reviewed the Petition for the words “mammal,” “mammalian,” “eukaryote,” and “eukaryotic” and could not locate any of them.

Linnarsson is not limited to mammalian cells. Nor is the Petition’s reliance on Linnarsson limited to mammalian cells.

ii. Not all of the mRNAs Present in a Cell must be Tagged

Petitioner argues that, “even assuming the claims required tagging mRNA from a mammalian cell, they do not require tagging **all** the mRNA molecules in such a cell.” Pet. Reply 12. Patent Owner concedes that the ’981 patent claims do not require tagging all of the polynucleotides (e.g., mRNA) but maintains that Petitioner’s obviousness challenge does. PO Sur-Reply 5; *see also* Tr. at 33:10–11 (“The claims don’t require this, but their theory does.”). Patent Owner’s lead argument in support of its characterization of the Petition states the following: “Petitioner argued POSITA would have been motivated to tag ‘**each**’ sample polynucleotide to resolve amplification bias.” PO Sur-Reply 5 (citing Pet. 53–55).

We reviewed pages 53 through 55 of the Petition for every occurrence of the quoted word “each.” It appears a total of four times over the cited pages. The first two occurrences are not relevant. *See* Pet. 54 (“each cell,” “each single cell”). The third and fourth occurrence both state “each *sample* polynucleotide.” *Id.* at 54–55.

During the hearing, counsel for Patent Owner acknowledged that the Petition proposed tagging each *sample* polynucleotide prior to amplification, but then immediately and confusingly suggested it does not.

MR. BAUGHMAN: . . . If you *look at Petition page 54* and, Your Honor, if you want to turn there I'll wait until the people pull it up. It's Paper 1, page 54 of the Petition.

JUDGE FITZPATRICK: I'm there. I'm there now.

MR. BAUGHMAN: Okay. Midway down the page on the right it says Linnarsson attempts to address the problem of amplification a certain way. Then at the bottom of that paragraph it says this approach only reduces amplification bias, it does not actually solve the problem. McCloskey, however, provides the exact solution. *And it's by incorporating a tag that uniquely identifies each sample polynucleotide prior to amplification. It doesn't say in the sample. It says each polynucleotide prior to amplification.* McCloskey solves it. That's their theory. That's what we responded to, and it is unfair to come up with some new theory now that we don't have an expert to talk about.

Tr. 35:3–16 (emphasis added).

It is clear enough to us that Petitioner argued that a skilled artisan would have modified Linnarsson in view of McCloskey in order to solve the problem of amplification bias, “[b]y incorporating a second tag in an oligonucleotide that uniquely identifies each *sample* polynucleotide prior to amplification,” and thus not every polynucleotide that might be present. Pet. 54 (emphasis added). Further, Dr. Cooper testified that a person of ordinary skill “could conceptually divide each of Linnarsson’s cell-tag sequence into two portions, wherein . . . the second portion performs the function of McCloskey’s ‘barcode’, *used to differentiate polynucleotides within a cell.*” Ex. 1002 ¶192 (emphasis added). Consistent with the Petition, Dr. Cooper does not state that *all* polynucleotides within a cell must be differentiated.

Neither the claims of the '981 patent nor Petitioner's rationale requires *all* of the mRNAs present in a cell to be tagged.

iii. McCloskey is not Limited to 16,384 Unique Barcodes

Petitioner argues that McCloskey's disclosure is not limited to 16,384 barcodes. Pet. Reply 13. Petitioner is correct, as McCloskey states the following:

Barcode

The number of distinguishable barcodes in a population of oligonucleotides used in a reaction is determined by the number of random bases, n . This enables one to distinguish among 4^n allele copies per reaction. We currently use seven nucleotides for the barcode, giving 16,384 possible barcodes for each encoding oligonucleotide.

Ex. 1004, 4. Not only is McCloskey not limited to 16,384 possible barcodes, it teaches precisely how to manipulate that number as needed. *Id.* In fact, McCloskey's authors/researchers stress that using seven nucleotides is what they "currently use." *Id.*

In its Sur-Reply, Patent Owner does not dispute that McCloskey teaches how to increase the number of possible barcodes. Instead, Patent Owner argues that the Petition relied on using seven nucleotides for the barcode and that using more nucleotides for the barcodes constitutes a "new argument" that "should have been 'presented earlier...to make out a *prima facie* case of unpatentability.'" PO Sur-Reply 6–7 (quoting *Juniper Networks, Inc. v. Swarm Tech. LLC*, No. IPR2021-01445, Paper 62, at 28–29 (PTAB Feb. 7, 2023)).

We disagree. Petitioner's argument that the barcode size could be increased as needed is based on McCloskey itself and is properly responsive to Patent Owner's argument that 16,384 unique barcodes would not be enough. Although the Petition clearly proposes using seven nucleotides, the Petition did not make any statement that it must be seven. Further, the very

reference Petitioner cites for the barcode feature, i.e., McCloskey, is clear that seven is not mandatory. Ex. 1004, 4. And, as Dr. Cooper's testimony makes clear, Linnarsson's native tag (which is conceptually divided into two portions per the modification) can be up to 20 nucleotides long and thus has sufficient space for a longer barcode. Ex. 1002 ¶192.

Neither McCloskey nor Petitioner's reliance on McCloskey is restricted to using barcodes that are seven nucleotides long.

In sum, we find Petitioner's rationale that a skilled artisan would have made the proposed combination to solve the problem of amplification bias persuasive.

9. Reasonable Expectation of Success

"[A]n obviousness determination requires not only the existence of a motivation to combine elements from different prior art references, but also that a skilled artisan would have perceived a reasonable expectation of success in making the invention via that combination." *Medichem*, 437 F.3d at 1165. "While the definition of 'reasonable expectation' is somewhat vague, [Federal Circuit] case law makes clear that it does not require a certainty of success." *Id.*

Petitioner argues that a person of ordinary skill in the art would have had a reasonable expectation of success in making the proposed modification because Linnarsson and McCloskey both teach adding oligonucleotides comprising a tag to polynucleotides, with the difference being that McCloskey's oligonucleotide comprises two tag sequences, whereas Linnarsson's has only one tag sequence. Pet. 55 (citing Ex. 1003, 19:5–10; Ex. 1004, 2); *see also* Ex. 1002 ¶192 (Dr. Cooper testifying that a

person of ordinary skill in the art “would have found this difference to be trivial”). Petitioner argues that the person of ordinary skill in the art “would have understood that Linnarsson’s tagged oligonucleotides—CDS or TSO—could be modified through routine methods to incorporate a second tag to uniquely identify each sample polynucleotide according to McCloskey’s teachings.” Pet. 55 (citing Ex. 1002 ¶¶192–193). According to Petitioner, the proposed modification “would be identical to the approach already set forth in Linnarsson, except that the tags in Linnarsson would be conceptually divided into two portions: one that codes for the unique cell and one that codes for the unique molecule.” *Id.* at 56 (citing Ex. 1002 ¶192). The cited testimony of Dr. Cooper indeed supports the argument. Ex. 1002 ¶92.

Patent Owner argues that there would not have been a reasonable expectation of success “of tagging ***each*** sample polynucleotide—*i.e.*, ***all*** the mRNA molecules in a sample—by incorporating McCloskey’s barcode into Linnarsson because McCloskey would not have provided enough unique barcodes to do so.” PO Resp. 40–41. Patent Owner also argues that there would not have been a reasonable expectation of success “of sequencing ‘the plurality of tagged nucleotides’ in [Petitioner’s] combination for the reasons described above in connection with the ‘sequencing’ limitation analysis.” *Id.* at 41. Neither of these arguments is persuasive because, as discussed above, these arguments are based on erroneous claim constructions that neither party has advanced.

Petitioner has demonstrated by a preponderance of the evidence that a person of ordinary skill in the art would have had a reasonable expectation

of success in combining the teachings of Linnarsson and McCloskey as proposed and resulting in the subject matter of claim 1.

10. Conclusion as to Claim 1

We conclude that claim 1 would have been obvious over Linnarsson and McCloskey as asserted by Petitioner based on the evidence presented.⁷

B. Claims 2–4 and 6

Claims 2, 3, 4, and 6 all depend from claim 1. Petitioner argues the proposed modification of Linnarsson in view of McCloskey, as already discussed above in relation to claim 1, includes each of the additional limitations recited in these dependent claims. Pet. 59–66. We address each claim below.

1. Claim 2

Claim 2 depends from claim 1 and recites “wherein the method further comprises amplifying the tagged polynucleotides prior to the sequencing step(c).” Ex. 1001, 30:49–51. For this limitation, Petitioner directs us to Linnarsson’s disclosure that tagged “cDNA samples are pooled and amplified” and then “sequencing is performed to obtain 100 million reads.” Ex. 1003, 4:11–12 (cited at Pet. 59). Patent Owner does not present any arguments specifically directed to claim 2, relying exclusively on its

⁷ “[E]vidence rising out of the so-called ‘secondary considerations’ must always when present be considered en route to a determination of obviousness.” *Stratoflex, Inc. v. Aeroquip Corp.*, 713 F.2d 1530, 1538 (Fed. Cir. 1983). The record before us, however, lacks any secondary considerations evidence.

dependence from claim 1. *See* PO Resp. 19–41 (arguing Ground 1 generally and not separately addressing claim 2).

We conclude that claim 2 would have been obvious over Linnarsson and McCloskey as asserted by Petitioner based on the evidence presented.

2. Claim 3

Claim 3 depends from claim 1 and recites “wherein the sample polynucleotides are selected from DNA and RNA.” Ex. 1001, 30:52–53. For this limitation, Petitioner directs us to Linnarsson’s disclosure of sample polynucleotides being mRNA and McCloskey’s disclosure of sample polynucleotides being DNA. Pet. 60–61 (citing Ex. 1003, 3:26–27; Ex. 1004, 1–2, 5). Patent Owner does not present any arguments specifically directed to claim 3, relying exclusively on its dependence from claim 1. *See* PO Resp. 19–41 (arguing Ground 1 generally and not separately addressing claim 3).

We conclude that claim 3 would have been obvious over Linnarsson and McCloskey as asserted by Petitioner based on the evidence presented.

3. Claim 4

Claim 4 depends from claim 3 and recites “wherein the sample polynucleotides comprise mRNA.” Ex. 1001, 30:53–54. For this limitation, Petitioner directs us to Linnarsson’s disclosure of sample polynucleotides being mRNA. Pet. 61 (referring back to analysis of claim 3). Patent Owner does not present any arguments specifically directed to claim 4, relying exclusively on its ultimate dependence from claim 1. *See* PO Resp. 19–41 (arguing Ground 1 generally and not separately addressing claim 4).

We conclude that claim 4 would have been obvious over Linnarsson and McCloskey as asserted by Petitioner based on the evidence presented.

4. *Claim 6*

Claim 6 depends from claim 1 and recites “wherein the tagged polynucleotides are generated through PCR or linear amplification of the plurality of sample polynucleotides using an adapter sequence comprising the MID sequence and an amplification primer.” Ex. 1001, 30:58–62. For this limitation, Petitioner first directs us to both references’ use of PCR in generating tagged polynucleotides. Pet. 62–63 (citing Ex. 1003, 10:18–22, 20:3–6; Ex. 1004, 2–3). Petitioner also directs us to both references’ use of “(1) an adapter sequence comprising the MID sequence and (2) an amplification primer molecule.” *Id.* at 63; *see also id.* at 63–66 (citing Ex. 1003, 19:18–23, 21:20–22, Figs. 3–4; Ex. 1004, 2–4; Ex. 1002 ¶¶228–229). Patent Owner does not present any arguments specifically directed to claim 6, relying exclusively on its dependence from claim 1. *See* PO Resp. 19–41 (arguing Ground 1 generally and not separately addressing claim 6).

We conclude that claim 6 would have been obvious over Linnarsson and McCloskey as asserted by Petitioner based on the evidence presented.

IV. Ground 2 (Claim 5)

Claim 5 depends from claim 1 and recites: “wherein the tagged polynucleotides are generated through at least one ligation reaction.” Ex. 1001, 30:56–57.

Petitioner argues that claim 5 would have been obvious over Linnarsson, McCloskey, and the knowledge of a person of ordinary skill in

the art because it would have been “a simple and obvious design choice with a reasonable expectation of success to introduce the[] tags by ligating an oligonucleotide comprising the tag sequences to Linnarsson’s CDS or TSO.” Pet. 66–67. This is so, according to Petitioner, because “it was well-known and routine in the art to attach tag sequences to other sequences by ligation.” *Id.* at 67 (citing Ex. 1002 ¶238). Petitioner additionally argues that the ordinarily skilled person “would have been motivated to utilize this design choice because it would allow for added flexibility in selecting tag sequences during an experiment.” *Id.* at 68 (citing Ex. 1002 ¶243). Petitioner argues that the ordinarily skilled person “would have had a reasonable expectation of success in ligating a second sequence to Linnarsson’s CDS or TSO” because ligating sequences to polynucleotides was “well-known and routine” in the art. *Id.* at 67.

A. Petitioner Adequately Articulated the Proposed Modification

Patent Owner argues that “Petitioner fails to make clear what is allegedly ligated to the CDS or TSO—a second tag only, or both first and second tags?” PO Resp. 42. Patent Owner argues that the Petition’s reference to multiple “sequences” being ligated, “apparently refer[s] to **both** the first and second tag sequences being ligated to the CDS or TSO.” *Id.* (quoting Pet. 67). Patent Owner argues that this is confusing because “Linnarsson’s CDS and TSO already have a first tag[], and the CDS and TSO of Ground 1 already (according to Petitioner) have first *and* second tags.” *Id.* (citing Pet. 47–49, 55–56, 67, 68).

We disagree that Petitioner failed to clearly articulate the proposed modification in Ground 2. The Petition asserts, without any ambiguity, that

the skilled artisan “would have had a reasonable expectation of success in ligating a second sequence to Linnarsson’s CDS or TSO.” Pet. 67. Dr. Cooper’s testimony in support is also clear.

Given that ligation was routine, conventional, and well-known in the art, it is my opinion that a POSA would also have had a reasonable expectation of success in ligating an additional sequence to Linnarsson’s CDS and/or TSO. For example, a POSA would have found it routine and obvious to ligate a tag sequence by using a linker molecule, such as the method described in Hug’s 2003 article.

Ex. 1002 ¶240 (citing Ex. 1007, 1). Further, the distinction between Ground 1 and Ground 2 is similarly clear.

As discussed [in Ground 1], a POSA would have been motivated to adopt McCloskey’s teachings and modify Linnarsson’s design so that the tags include a first region, which could distinguish between different cells, and a second region, which could distinguish tagged polynucleotides originating from the same single-cell. Further, it would have been an obvious and a routine design choice with a reasonable expectation of success to carry out this modification by ligating an oligonucleotide comprising a second tag sequence to Linnarsson’s CDS and/or TSO.

Ex. 1002 ¶237.

Many of Patent Owner’s arguments are premised on a different modification than that which Petitioner proposes. *See, e.g.*, PO Resp. 47 (“Ppetitioner ignores that, per Petitioner’s Ground 1 arguments, its modified CDS and/or TSO from Ground 1 *already contain(s) the two tags.*[] . POSITA would not have been motivated to add *more* tags (and more steps) when the CDS or TSO (according to Petitioner’s theory) already contain two tags.”), 47 (“To the extent Petitioner’s theory is that POSITA would have further modified its Ground 1 combination to **remove** the tag(s) from the

Linnarsson/McCloskey CDS and/or TSO, ‘instead’ ligating the tags to the CDS or TSO after extension, that, too, fails.” (quoting Pet. 68)), 48 (“Petitioner’s theory fails because removing allegedly-existing tags and then ligating new tags to Ground 1’s Linnarsson-based method would fundamentally alter its principle of operation.”). All such arguments are not persuasive as irrelevant.

B. Petitioner Articulated Adequate Reasons to Support the Proposed Modification

Patent Owner argues that “Petitioner’s sole purported justification for th[e] proposed modification is the conclusory assertion that ‘it would allow for added flexibility in selecting tag sequences during an experiment’.” PO Resp. 50 (quoting Pet. 68). That is not accurate. Petitioner also argues that ligating a second sequence to Linnarsson’s native CDS and/or TSO would have been “a simple and obvious design choice” and involve only “well-known and routine” methods. Pet. 66–7. In fact, as pointed out by Petitioner (see Pet. Reply 18), the ’981 patent acknowledges that ligation was routine and conventional in the art, stating:

The practice of the present invention may employ, unless otherwise indicated, conventional techniques and descriptions of organic chemistry, polymer technology, molecular biology (including recombinant techniques), cell biology, biochemistry, and immunology, which are within the skill of the art. *Such conventional techniques include* polymer array synthesis, hybridization, *ligation*, and detection of hybridization using a label.

Ex. 1001, 13:32–39 (emphasis added); see also Ex. 1002 ¶238 (quoting the same).

Patent Owner does not show that the proposed of modification would have been beyond the level of skill in the art. Nor does Patent Owner show that the modification would result in anything beyond what would have been predicted based on these prior art teachings. In fact, the Patent Owner Response does not even address Petitioner's design choice rationale. PO Resp. 46–52. Patent Owner's declarant, Dr. Quackenbush does “not contest[] that ligation was a well-known method of attaching oligonucleotides.” Ex. 1057, 187:23–25; *see also id. at* 188:17–8 (“I told you earlier ligation was a conventional technique.”).⁸

As discussed above, Petitioner additionally argues that a skilled artisan would have ligated the second sequence to the CDS and/or TSO for added flexibility in selecting tag sequences during an experiment. Pet. 68. Petitioner argues “this approach would be entirely equivalent to the method

⁸ More broadly, during his cross-examination, Dr. Quackenbush could not describe *anything* he thought was “inventive” about the claims of the '981 patent, stating as follows: “I haven't performed that analysis, so I would be hesitant to render an opinion today since it's outside the scope of my report” (i.e., declaration). Ex. 1057, 203:6–204:5. Yet, in his declaration, Dr. Quackenbush implies a familiarity with the “invention” of the '981 patent, including a section titled “The '981 Patent's Invention.” Ex. 2013 ¶¶27–32. In that section, he testifies: “*Until inventions such as the '981 Patent*, most nucleotide sequencing was performed using polynucleotides (e.g., DNA or RNA) from a population of cells in a process called bulk analysis.” *Id.* ¶29 (emphasis added). Near the end of his declaration, Dr. Quackenbush “conclude[s] that each of the Challenged Claims is not invalid as obvious under 35 U.S.C. § 103.” *Id.* ¶107. The inventiveness (or not) of the claims of the '981 patent was squarely within the scope of Dr. Quackenbush's direct testimony. His refusal or inability, on cross-examination, to identify anything inventive about the claims reinforces our ultimate conclusion that they are unpatentable under 35 U.S.C. § 103.

already disclosed by the combination of Linnar[s]son and McCloskey [applied in Ground 1], except that instead of having the tag sequence be part of the CDS or TSO prior to extension, the tag sequences are added to the CDS or TSO after extension.” *Id.* (citing Ex. 1002 ¶243).

Patent Owner argues that this rationale is unclear and conclusory. PO Resp. 50. However, we find this rationale sufficiently clear and not in need of further explanation. It is clear from the Petition that the “flexibility” is the option to add a second tag sequence later in the process and to possibly tailor its length (i.e., number of nucleotides) based on specific needs. Pet. 68 (citing Ex. 1002 ¶243). In the testimony cited by the Petition, Dr. Cooper explains:

A POSA would have understood that such an approach, using ligation to add the tag sequences to the CDS and/or TSO, would have been entirely equivalent to the method disclosed by the combination of Linnarsson and McCloskey. The only difference is in the timing: with the combination of Linnarsson and McCloskey discussed [in Ground 1], the tag sequence is part of the CDS or TSO prior to extension whereas by using ligation, the tag sequence is added after extension. *Using ligation to add the tag after extension would allow for increased flexibility in selecting tag sequences during an experiment.* Thus, a POSA would have understood that using ligation was a routine choice and would have been motivated to use ligation because of the increased flexibility in tag choice.

Ex. 1002 ¶243 (emphasis added).

Patent Owner also argues that “Petitioner’s assertion that ligation ‘would be entirely equivalent to the method already disclosed by Linnarsson’ negates any motivation to modify.” PO Resp. 51 (quoting Pet. 68; citing *In re Schweickert*, 676 F. App’x 988, 995 (Fed. Cir. 2017)).

Patent Owner adds that its position “is especially true where, as here, the modification would add more steps and complexity to achieve only an ‘equivalent’ method.” *Id.* (quoting Pet. 68; citing *Weber, Inc. v. Provisur Techs., Inc.*, No. IPR2019-01461, Paper 36, at 42–43 (Feb. 10, 2021)).

We are not persuaded by this argument nor by the cases Patent Owner cites. First, that a known technique or feature is equivalent to a limitation of a claim supports, not negates, the obviousness of substituting that technique or feature. *See In re Fout*, 675 F.2d 297, 301 (CCPA 1982) (“Because both [references] teach a method for separating caffeine from oil, it would have been *prima facie* obvious to substitute one method for the other. Express suggestion to substitute one equivalent for another need not be present to render such substitution obvious.”). Further, “just because better alternatives exist in the prior art does not mean that an inferior combination is inapt for obviousness purposes.” *In re Mouttet*, 686 F.3d 1322, 1334 (Fed. Cir. 2012).

What Patent Owner seems to be arguing is: why would a person of ordinary skill in the art substitute an equivalent technique or step (here, ligation) if it creates more complexity or work? The answer to that question is the flexibility that Petitioner identifies. Petitioner’s identification of that benefit renders Patent Owner’s reliance on *Schweickert* and *Weber* inapposite.

In *Schweickert*, the Court vacated a Board decision concluding that application claims would have been obvious over two references. *Schweickert*, 676 F. App’x at 990. The Court vacated the Board’s decision for multiple reasons. *See id.* at 995 (“The Board’s decision is deficient in another respect. . . . We note one more shortcoming in the Board’s

decision.”). One such reason was that the Board’s obviousness conclusion was based on a modification of a random-access-memory component that would “add unwanted cost and complexity to the system” *without explaining any benefit of doing so*. See *id.* (“Although [the Board] asserts that Birrell would have ‘benefitted from the advantages of . . . Cunniff’s semaphore mechanism,’ the Board did not explain why this is so.”).

In *Weber*, as argued by Patent Owner (see PO Resp. 51), the Board concluded that a person of ordinary skill in the art would not have carried out the particular combination asserted by the petitioner because it would have introduced “needless complexity.” *Weber*, No. IPR2019-01461, Paper 36, at 43. *Weber* is silent, however, as to whether the petitioner had identified a benefit despite introducing “needless complexity.” *Id.* at 42–43. If anything, the Board’s labeling the complexity “needless” suggests that the record lacked any identified benefit. *Id.* at 43.

We find Petitioner’s rationale that a skilled artisan would have made the proposed combination because it was an obvious design choice based on the well-known and routine technique of ligating sequences to polynucleotides as well Petitioner’s identification of increased flexibility in selecting tag sequences during an experiment sufficient to support the proposed modification.

C. Petitioner Has Demonstrated a Reasonable Expectation of Success

Petitioner argues that a person of ordinary skill in the art would have had a reasonable expectation of success in carrying out the proposed modification because ligating tags to polynucleotides was well-known and routine. Pet. 67. Petitioner argues that it could have been done, for

example, by using a linker molecule, as such a method was known in the prior art. *Id.* at 67–68 (Ex. 1002 ¶¶240–243; Ex. 1007, 7–8; Ex. 1024, 5; Ex. 1039, 1–2, Fig. 1; Ex. 1040 ¶¶71–75, 235–239, Fig. 2). The evidence Petitioner cites in support is Dr. Cooper’s testimony (Ex. 1002) along with four references that are not Linnarsson or McCloskey (Ex. 1007; Ex. 1024; Ex. 1039; Ex. 1040). Petitioner describes one of the four references in particular, arguing as follows: “A 2003 article (“Hug” [(Ex. 1007)]) discloses ligating tag sequences to a polynucleotide by hybridizing a linker to the target polynucleotide, which is designed to hybridize to another polynucleotide used as a MID such that the MID is then ligated to the target polynucleotide.” Pet. 67–68 (citing Ex. 1007, 8).

Patent Owner argues that Petitioner’s reliance on Hug is “improper” and that a person of ordinary skill in the art could not ligate a second sequence onto Linnarsson’s CDS or TSO and achieve the subject matter of claim 5. PO Resp. 46, 53–61.

Petitioner’s reliance on Hug and other references is not improper. Petitioner’s Ground 2 asserts that claim 5 would have been obvious over “Linnarsson in View of McCloskey, and *Further in View of the Knowledge of a POSA.*” Pet. 66 (emphasis added). Petitioner cited four references to support its argument that a person of ordinary skill in the art would have found ligating a sequence to a polynucleotide well-known and routine. Pet. 67–68 (citing Ex. 1007, 7–8; Ex. 1024, 5; Ex. 1039, 1–2, Fig. 1; Ex. 1040 ¶¶71–75, 235–239, Fig. 2); *see also* Ex. 1002 ¶¶240–243 (Dr. Cooper testifying that ligation was “well-known, conventional, and routine” in the art); Ex. 1057, 187:23–25, 188:17–8 (Dr. Quackenbush agreeing that ligation was “well-known” and “conventional” in the art).

Patent Owner argues that “the biotin on the CDS’s 5’ end would block ligation of tag(s) to the CDS.” PO Resp. 53.⁹ Patent Owner argues that, “[o]nce a biotin molecule is covalently bound to the end of these nucleic acids, it prevents any other molecule, including a nucleic acid tag, from being covalently bound to the end of the nucleic acid to which the biotin is already bound.” *Id.* at 55 (citing Ex. 2013 ¶89). Patent Owner’s argument is premised on Linnarsson requiring its CDS to be “capped by a ‘BIO’ (biotin) molecule on its 5’ end.” *Id.* at 53 (citing Ex. 1003, Fig. 11).

Petitioner replies that Linnarsson’s disclosure is not limited to Figure 11 and does not require biotin. Pet. Reply. 23 (citing Ex. 1003, Figs. 3–4). Petitioner also points out that, in describing Figure 11, Linnarsson states that it “show[s] graphical representation of steps ***optionally used*** in the STRT method.” *Id.* (quoting Ex. 1003, 5:28–29, with emphasis added by Petitioner). Petitioner also argues that, if a person of ordinary skill in the art wanted to use biotin, she would know “the biotin could not be at the end of the CDS or TSO but would instead need to be added after ligation or included with the barcode.” Pet. Reply 24 (citing Ex. 1059 ¶¶109–112).

Patent Owner argues that Petitioner’s argument that Linnarsson does not require biotin is a new and improper argument on reply. PO Sur-Reply 19. We disagree. Petitioner’s argument is directly responsive to Patent Owner’s argument. See 37 C.F.R. § 42.23(b) (“A reply may only respond to

⁹ Patent Owner also argues that there would be complications in ligating a second tag or sequence to Linnarsson’s TSO. Because we find, as discussed below, there would be a reasonable expectation of success in ligating a second tag or sequence to Linnarsson’s CDS, we need not and do not reach Patent Owner’s TSO-based arguments.

arguments raised in the corresponding opposition, patent owner preliminary response, patent owner response, or decision on institution.”). Petitioner’s argument is not only proper, but also persuasive; Linnarsson does not require biotin. *See* Ex. 1003, 3:25–4:2 (“Summary of the Invention” not mentioning biotin), 5:28–29 (“Figure 11 [which does include biotin], panels A-F, show graphical representation of steps *optionally used* in the STRT method.” (emphasis added)), Figs. 3–4 (not including biotin).

Patent Owner additionally argues that “Petitioner has not explained how Linnarsson’s method would work without biotin.” PO Sur-Reply 19. However, Petitioner bears no burden in that regard. *See Apple Inc. v. Corephotonics, Ltd.*, 861 F. App’x 443, 450 (Fed. Cir. 2021) (holding that in the context of AIA trial proceedings, “regardless of the forum, prior art patents and publications enjoy a presumption of enablement, and the patentee/applicant has the burden to prove nonenablement for such prior art” and that “[i]t was error for the Board to suggest otherwise”).

Further, we are persuaded by Petitioner’s argument that a person of ordinary skill in the art would have known how to use biotin in a manner that did not interfere with the proposed modification. Pet Reply 24 (citing Ex. 1059 ¶¶109–112); *see also KSR*, 550 U.S. at 421 (“A person of ordinary skill is also a person of ordinary creativity, not an automaton.”); Ex. 1057, 175:2–6 (Dr. Quackenbush: “So if there’s already a biotin attached, you wouldn’t be able to create that bond, hence, ligation wouldn’t occur. And a person in 2009 would have understood this.”).

In sum, we find that a person of ordinary skill in the art would have had a reasonable expectation of success in the proposed modification.

III. CONCLUSION¹⁰

We conclude, based on a preponderance of the evidence, that claims 1–6 are unpatentable. In summary:

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not Shown Unpatentable
1–4, 6	103(a)	Linnarsson, McCloskey	1–4, 6	
5	103(a)	Linnarsson, McCloskey, “further in view of the knowledge of a [person of ordinary skill in the art]”	5	
Overall Outcome			1–6	

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that claims 1–6 of U.S. Patent No. 10,155,981 B2 are proven unpatentable;

¹⁰ Should Patent Owner wish to pursue amendment of the challenged claims in a reissue or reexamination proceeding subsequent to the issuance of this decision, we draw Patent Owner’s attention to the April 2019 *Notice Regarding Options for Amendments by Patent Owner Through Reissue or Reexamination During a Pending AIA Trial Proceeding*. See 84 Fed. Reg. 16,654 (Apr. 22, 2019). If Patent Owner chooses to file a reissue application or a request for reexamination of the challenged patent, we remind Patent Owner of its continuing obligation to notify the Board of any such related matters in updated mandatory notices. See 37 C.F.R. § 42.8(a)(3), (b)(2).

FURTHER ORDERED that Patent Owner's Motion To Exclude is granted-in-part and denied-in-part;

FURTHER ORDER that Petitioner's Motion To Exclude is dismissed as moot; and

FURTHER ORDERED that, because this Decision is final, a party to the proceeding seeking judicial review of the Decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2023-00876
Patent 10,155,981 B2

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